

VENKAT KOUSHIK PILLALA

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Portfolio | LinkedIn | GitHub

SUMMARY

AI/ML Engineer with an M.S. in Applied Artificial Intelligence from Stevens Institute of Technology, focused on LLM evaluation, RAG systems, NLP, and applied AI deployment. Experienced in building live AI demos, retrieval workflows, model evaluation pipelines, and cloud-based ML systems using Python, PyTorch, Hugging Face, FastAPI, Docker, and AWS.

TECHNICAL SKILLS

Programming & APIs: Python, SQL, C++, FastAPI, REST APIs, Linux/Bash, Git

Machine Learning & NLP: PyTorch, Hugging Face Transformers, Scikit-learn, TensorFlow, XGBoost, NLP, Text Classification, Computer Vision, Model Evaluation

Generative AI & Retrieval: LLMs, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LoRA Fine-Tuning, Prompt Engineering, LLM Evaluation, Embeddings, Reranking

Vector Databases & Data Systems: FAISS, ChromaDB, LanceDB, PostgreSQL, MongoDB, BigQuery, Apache Spark

Cloud & MLOps: AWS SageMaker, Amazon S3, AWS Glue, AWS Lambda, Amazon EC2, Amazon Rekognition, Amazon Comprehend, Docker, MLflow, CI/CD, Model Deployment

PROFESSIONAL EXPERIENCE

Aspire Technolab | *ML Engineer Intern*

Jan. 2024 – May 2024

Python, PyTorch, Hugging Face Transformers, LLaMA 2, LoRA, NLP, OpenCV, CNNs, REST APIs

- Contributed to a team-led enterprise AI initiative exploring domain-specific LLM workflows with LLaMA 2 and LoRA for internal use-case validation.
- Handled data preparation, tokenization, prompt testing, benchmark runs, and output review to evaluate model relevance and response quality.
- Developed NLP classification components for sentiment analysis and spam detection using PyTorch and Hugging Face Transformers.
- Achieved a **0.92 F1-score** through preprocessing improvements, precision/recall analysis, confusion-matrix review, and recurring error analysis.

FunMotionLabs | *Game Development Intern, AI/ML Systems*

Jan. 2023 – Dec. 2023

Python, NLP, Machine Learning, Reinforcement Learning, Unity, C Sharp, FSM

- Designed AI-driven agent behavior using machine learning, heuristic decision logic, and Finite State Machines for adaptive real-time interactions.
- Implemented NLP-based interaction workflows to interpret user inputs, identify intent triggers, and map them to context-aware agent responses.
- Applied reinforcement learning concepts and simulation-based testing to improve navigation behavior, response consistency, and decision selection.
- Delivered reusable AI behavior components for the Kurenai module, supporting integration into the broader interactive application workflow.

Credo Systemz | *AWS ML Developer Trainee*

July 2022 – Oct. 2022

AWS SageMaker, Amazon S3, AWS Glue, AWS Lambda, EC2, Docker

- Implemented AWS SageMaker workflows covering S3 data ingestion, feature preparation, model training, evaluation, and endpoint deployment.
- Developed ETL and feature engineering pipelines using AWS Glue and SageMaker Data Wrangler for structured ML training inputs.
- Deployed inference workflows using SageMaker endpoints, AWS Lambda, EC2, Docker, and REST APIs for scalable model testing.
- Built applied AI workflows using Amazon Rekognition for image analysis and Amazon Comprehend for NLP-based entity extraction.

PROJECTS

Secure Local RAG and LLM Evaluation Platform | Live Demo

Feb. 2026 – Apr. 2026

Python, LLaMA 3, LangChain, ChromaDB, FastAPI, Docker, SQL

- Built a privacy-focused local RAG platform for semantic search and question answering over sensitive documents and codebases without external API reliance.
- Implemented ingestion, vector retrieval, FastAPI inference, SQL-backed evaluation, and metadata tracking for repeatable local AI testing.
- Reduced hallucinated responses by **34%** through retrieval tuning, answer review, failure analysis, and evaluation-driven prompt refinement.

LLM Code Review Assistant | Live Demo

April 2026 – May 2026

Python, FastAPI, Streamlit, LangGraph, Static Analysis, SQLite, Docker

- Developed an AI-assisted code review system combining static analysis, security checks, scoring, and LLM-based reasoning.
- Designed review workflows to flag maintainability issues, risky coding patterns, missing best practices, and improvement opportunities.
- Deployed a live Hugging Face demo with a Streamlit interface for interactive Python code review and structured reviewer-style feedback.

LLM Red Team Risk Evaluator | Live Demo

May 2026 – June 2026

Python, Streamlit, LLM Evaluation, Prompt Testing, Risk Scoring, Safety Evaluation

- Built an LLM evaluation tool for testing adversarial prompts, jailbreak attempts, hallucination risk, refusal behavior, and unsafe response patterns.
- Created structured risk-scoring workflows to assess response safety, reliability, prompt sensitivity, and model behavior across red-team scenarios.
- Deployed an interactive Hugging Face demo to present prompt-level risk analysis through a clear evaluation dashboard.

EDUCATION

Stevens Institute of Technology

Hoboken, NJ

Master of Science in Applied Artificial Intelligence | **CGPA: 3.76**

August 2024 – May 2026

Relevant Coursework: Machine Learning; Deep Learning; Big Data Analytics; Pattern Recognition & Classification; Probability & Stochastics; Data Structures & Algorithms

Sathyabama University

Chennai, India

B.E. in Computer Science Engineering | **CGPA: 8.7**

August 2020 – May 2024

Relevant Coursework: Data Structures; Algorithms; Database Management Systems; Operating Systems; Computer Networks; Data Science; Business Analytics

LEADERSHIP & INVOLVEMENT

- Led AWS Cloud Club and AI-UI/UX Club activities focused on cloud learning, AI project discussions, product design, and student collaboration.
- Coordinated Debate Club sessions, strengthening public speaking and structured communication.